Tonal noise of a ducted desk fan with serrated blades: sliding-mesh URANS, Ffowcs Williams–Hawkings prediction, and a force-based verification of the acoustic output

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Abstract

A computational-aeroacoustics chain was set up for a five-blade ducted desk fan with owl-inspired serrated trailing edges. The incompressible flow is solved with OpenFOAM (URANS, k–ω SST) on a 1.95-million-cell mesh with a cyclicAMI rotor zone at 1 509 rpm, and the Farassat 1A form of the Ffowcs Williams–Hawkings (FW-H) equation gives the pressure at four microphones 1 m away. The rotor produces 43.3 mN of thrust and 108 m^3/h. The spectra contain the blade-passing frequency (BPF, 126 Hz) 20 to 28 dB above the local floor, a once-per-revolution tone confined to the rotor plane whose directivity matches the rotating in-plane force found in the blade loads, and a comb of lines spaced 100.5 Hz above 600 Hz that follows periodic bursts in the blade force. A verification of the FW-H output against compact-dipole estimates built from the integrated pressure force reproduces the directivities but not the amplitudes (-9, $+14$ and -18 dB for the mean load, the 25 Hz tone and the BPF), which points to an input or scaling problem in the acoustic step. The results are therefore the tone frequencies and directivities, not the levels. No mesh or time-step independence study was run, and the effect of the serrations is not measured.

1 Introduction

The tonal noise of a low-speed fan is radiated by the unsteady force the blades exert on the air. A rotor of Z identical blades turning at n revolutions per second radiates at the blade-passing frequency Zn and its harmonics [1]; interaction with V stationary struts adds circumferential modes $m = kZ - sV$ [2], and any asymmetry between blades adds tones at multiples of n. Trailing-edge serrations, inspired by owl feathers, act mainly on broadband noise by reducing the scattering of boundary-layer turbulence at the edge [3, 4]. A recent study combined owl-feather serrations with cicada-wing geometry on small propellers and reported lower overall levels and higher propulsive efficiency [5]; it was the starting point of the fan studied here.

The question was practical: can an open-source chain, run on a small self-built cluster, give a usable acoustic signature of a 3D-printable desk fan before it is built, and which parts of that signature can be trusted? The fan has five serrated blades in a short duct, with a motor hub held by three arms. The contributions are:

- a sliding-mesh URANS and FW-H set-up for the fan (OpenFOAM, libAcoustics) that runs stably;

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- a per-record spectral analysis of the FW-H signals that identifies the BPF and its harmonics, a once-per-revolution tone explained by the force history, and a high-frequency line comb tied to periodic bursts in the blade force;
- a verification of the FW-H amplitudes against compact-dipole estimates built from the integrated blade pressure force, which separates the reliable part of the acoustic output (frequencies, directivities) from the unreliable part (levels).

2 Related work

The Ffowcs Williams–Hawkings equation [6] extends Lighthill's analogy to sources on moving surfaces. For surfaces moving at subsonic speed, Farassat's Formulation 1A gives the thickness and loading noise as retarded-time surface integrals with the observer-time derivative taken analytically [7]; it is the standard tool for rotor tonal noise [8]. Tao et al. [9], in a preprint, argue that Formulation 1A omits a term linked to the variation of the surface normal. The field of a point force in motion, which underlies the compact-dipole estimates of Section 5, was derived by Lowson [1], and the modal theory of rotor–stator interaction by Tyler and Sofrin [2]. The FW-H implementation used is libAcoustics [10, 11], a library of function objects for OpenFOAM [12, 13]. Rotor motion uses the arbitrary mesh interface, whose conservative interpolation follows the local Galerkin projection of Farrell and Maddison [14], and the flow uses the k–ω SST model [15]. On serrations, Howe [3] gave the first model of sawtooth trailing-edge noise and Lyu et al. [4] a more complete prediction; both concern broadband noise, which URANS does not resolve. Spectral estimation follows Harris [16] for windowing and Welch [17] for segment averaging.

3 Model

3.1 Geometry and operating point

The geometry is the Polytechnia FreeCAD model of the fan (Table 1): five-blade rotor (tip radius 69.2 mm, serrated trailing edges), duct of inner radius 76 mm, motor hub and three support arms. The rotor turns at $\Omega = 158\,\mathrm{rad/s}$ (1 509 rpm, $n = 25.15\,\mathrm{Hz}$), so BPF $= 5n = 125.7\,\mathrm{Hz}$. The tip speed is 10.9 m/s (Mach 0.03): the flow is incompressible and the rotor is acoustically compact at the BPF ($kR = 0.16$).

Table 1: Model parameters (files in `openfoam_case/`).

Quantity	Value
Rotor	5 blades, tip radius 69.2 mm, serrated trailing edges
Rotation	$\Omega = 158\,\mathrm{rad/s}$ (1 509 rpm), ramp from 0 over 5 ms
Fluid	air, $\rho = 1.2\,\mathrm{kg/m^3}$, $\nu = 1.5 \times 10^{-5}\,\mathrm{m^2/s}$
Domain	box $0.4\,\mathrm{m} \times 0.4\,\mathrm{m} \times 0.5\,\mathrm{m}$, axis along $+z$
Open boundaries	total pressure $p_0 = 0$, `pressureInletOutletVelocity`
Walls	no slip; blades `movingWallVelocity`; wall functions for k, ω
Inflow turbulence	$k = 0.54\,\mathrm{m^2/s^2}$, $\omega = 128\,\mathrm{s^{-1}}$
Observers	$r = 1\,\mathrm{m}$: downstream and upstream on the axis, radial, 45° downstream

3.2 Equations

The incompressible Navier–Stokes equations are solved with the k–ω SST closure. The rotor zone turns as a rigid body; fluxes cross the cylindrical interface ($R = 72\,\mathrm{mm}$) through the AMI. The acoustic pressure at observer $\mathbf{x}$ is obtained from the FW-H equation with the blade skin as integration surface, keeping the thickness and loading terms and neglecting the quadrupole term:

$$4\pi p'_T(\mathbf{x},t) = \int_S \left[\frac{\rho_0(\dot{v}_n + v_{\dot{n}})}{r(1-M_r)^2}\right]_{\mathrm{ret}} \mathrm{d}S + \int_S \left[\frac{\rho_0 v_n\,(r\dot{M}_r + c_0 M_r - c_0 M^2)}{r^2(1-M_r)^3}\right]_{\mathrm{ret}} \mathrm{d}S, \tag{1}$$

$$4\pi p'_L(\mathbf{x},t) = \frac{1}{c_0}\int_S \left[\frac{\dot{\ell}_r}{r(1-M_r)^2}\right]_{\mathrm{ret}} \mathrm{d}S + \int_S \left[\frac{\ell_r - \ell_M}{r^2(1-M_r)^2}\right]_{\mathrm{ret}} \mathrm{d}S + \frac{1}{c_0}\int_S \left[\frac{\ell_r\,(r\dot{M}_r + c_0 M_r - c_0 M^2)}{r^2(1-M_r)^3}\right]_{\mathrm{ret}} \mathrm{d}S, \tag{2}$$

with $p' = p'_T + p'_L$, in the notation of Farassat [7] and Brentner and Farassat [8]: v_n is the normal surface velocity, ℓ_r and ℓ_M are the surface loading projected on the radiation direction and on the surface Mach number vector, M_r is the Mach number in the radiation direction, and dots denote source-time derivatives. The solver works with a kinematic pressure; the FW-H input uses $\rho_0 = 1.2\,\mathrm{kg/m^3}$ and $c_0 = 343\,\mathrm{m/s}$.

4 Numerical set-up

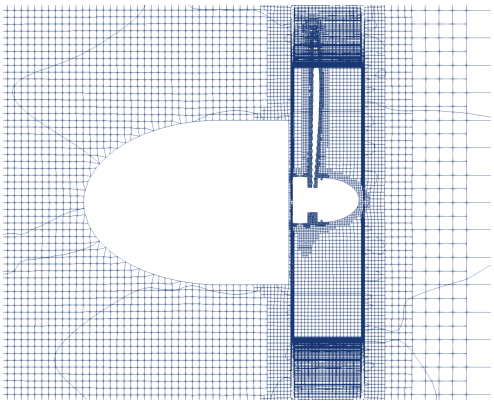
4.1 Mesh

The mesh is built with `snappyHexMesh` from a 10 mm background grid, refined to level 5 (0.31 mm) on the blade surfaces so that the serrations are resolved, with refined rotor and wake cylinders (Figure 1). The STL surfaces of this mesh were extracted from the boundary of an earlier mesh, so they are faceted. The faceZone of the rotor cylinder is split into a cyclicAMI pair with `createBaffles`. The retained mesh has 1 949 584 cells (Table 2); no prism layers were inserted, and the blade y^+ averages about 4 and reaches 445.

The data trail needs one remark. The mesh dictionary of the 1.95 M-cell mesh (now `openfoam_case/system/snappyHexMeshDict`) matches the mesh and quality figures above. An earlier run of the same case on a 906 936-cell mesh (blades at level 4, dictionary kept as `snappyHexMeshDict.baseline_0p91M`) was computed on a desktop under WSL with OpenFOAM v2406 from 15 to 18 June 2026 and reached 0.276 s; its FW-H output differs from the published records and is not used. The published records (written 25–27 June) come from the cluster runs with OpenFOAM v2506; the cluster has been offline since, so the link between these records and the 1.95 M-cell mesh could not be re-checked from the case directories.

4.2 Solver, time strategy and computing

`pimpleFoam` is run with three outer correctors, two pressure correctors and one non-orthogonal corrector. Momentum convection uses `linearUpwindV`, k and ω use `limitedLinear`, and time integration is first-order implicit Euler. Phase 1, from rest to 0.19 s (4.8 revolutions), uses an adaptive time step with a Courant limit of 3 and $\Delta t \le 1\times10^{-4}\,\mathrm{s}$. Phase 2 restarts with a fixed $\Delta t = 3.5\times10^{-5}\,\mathrm{s}$ (227 steps per BPF period) and the FW-H function object, up to 0.39 s; the fixed step is confirmed by the force output, written every 10 steps at a uniform $3.5\times10^{-4}\,\mathrm{s}$. The FW-H



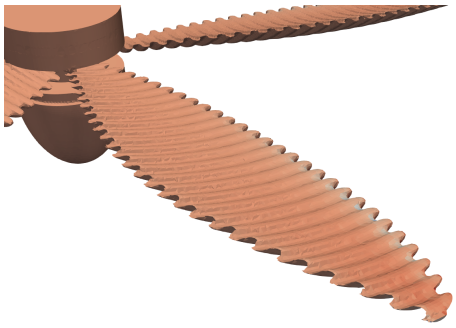


Figure 1: Left: meridional cut of the mesh, from the far field to the blade band and the AMI cylinder ($R = 72\,\mathrm{mm}$). Right: serrated trailing edges as resolved by the mesh.

Table 2: `checkMesh` on the retained mesh (June 2026 report).

Cells / points	1 949 584 / 2 205 401
Max aspect ratio	6.61
Max / mean non-orthogonality	61.4 / 8.2
Max skewness	7.25 (12 faces above 4)
Concave cells	49 827 (2.6 %)
Cells with determinant < 0.001	1851
AMI weights (min / mean / max)	0.90 / 1.002 / 1.27

pressure is sampled every 5 steps ($f_s = 5\,714\,\mathrm{Hz}$) in two records of 100 ms. The solver runs only with floating-point trapping off (`FOAM_SIGFPE=false`) because of the degenerate cells.

Stable transient runs required three changes. A first attempt with the non-conformal coupling of OpenFOAM.org stopped in the mesh stitcher after about 200 µs on six mesh variants, so the case moved to cyclicAMI on the ESI release; `snappyHexMesh` has to run in serial, because the parallel run aborted while load-balancing the AMI faceZone; and the pressure solver was switched from GAMG to PCG on the coarse test mesh. The runs used a two-node Linux cluster built for this project (8 and 4 cores, MPI over Ethernet); one node saturates at 4 cores, and splitting the AMI case across both nodes gave no gain, so the nodes ran independent cases.

4.3 Acoustic post-processing

Each FW-H record begins with a start-up transient (on the axial microphones its peak reaches 35 to 52 robust standard deviations of the rest of the record). The first 15 ms of each record are discarded, which leaves two segments of 85 ms (486 samples) separated by a 15 ms gap. Each segment has its mean and linear trend removed, is Hann-windowed and zero-padded to 8192 points, and the two power spectra are averaged; the true resolution is 11.8 Hz. Tone levels are the peaks within ± 6 Hz of n and of the BPF, and the local floor is the median level between 140 and 240 Hz. Appendix A explains why the segments are not concatenated and no filter is applied.

5 Verification

Established regime. The force record available here starts at a restart at 0.151 s (Figure 2). Averaged over successive revolutions, the thrust changes by 1.9 % between the revolutions starting at 3.8 and 4.8 rotor turns, and by 0.7 % between 4.8 and 5.8 turns.

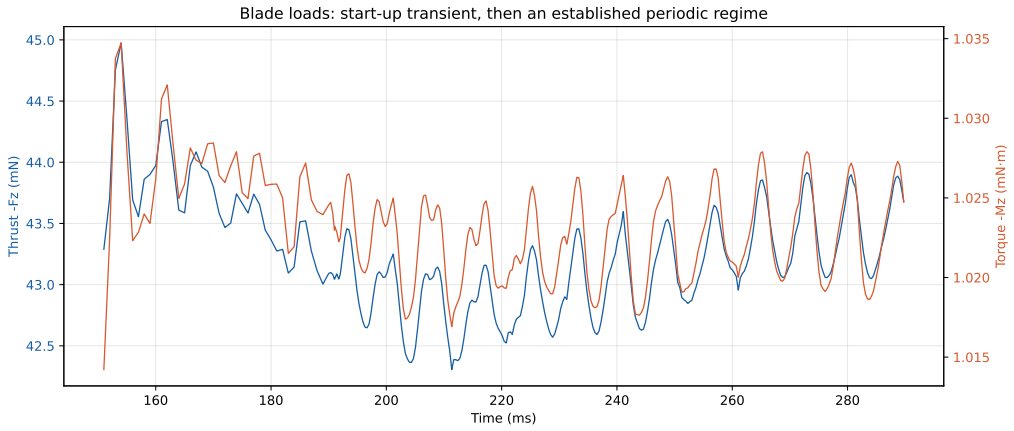


Figure 2: Thrust and torque on the blades from the restart at 0.151 s: the jump at the restart, then a periodic regime.

Coherence between records. On the radial and 45° microphones, a sine fit at 25.15 Hz on the absolute time axis gives the same level and the same phase (148°) in both segments. On the axial microphones the fitted phase changes between segments and the level stays at the floor.

Force-based check of the FW-H amplitudes. At 0 Hz, 25 Hz (wavelength 13.6 m) and the BPF (2.7 m) the rotor is compact, so its loading noise reduces to that of the net force on the blades applied at the hub [1]:

$$p'(\mathbf{x},t) = \frac{1}{4\pi}\left[\frac{x_i}{c_0 r^2}\frac{\mathrm{d}F_i}{\mathrm{d}t} + \frac{x_i F_i}{r^3}\right]_{t-r/c_0}, \qquad |p'| = \frac{A}{4\pi r}\sqrt{\left(\frac{\omega}{c_0}\right)^2 + \frac{1}{r^2}} \tag{3}$$

for a harmonic component of amplitude A along the observer direction (the near-field term dominates at 25 Hz, where $kr = 0.46$). Because the FW-H surface integrates the blade pressure only, F is the pressure part of the force output. Table 3 compares the estimates with the FW-H output over the same time window. The directivities agree: the mean signal is +1.24, −1.11, +0.86 and ≈0 mPa on the downstream, upstream, 45° and radial microphones, the $\cos\theta$ pattern of the axial force, and the 25 Hz tone is absent on the axis and 3.1 dB weaker at 45° than in the plane ($\cos 45°$ gives 3.0 dB). The amplitudes do not. The steady case is the simplest: it involves no time derivative and no retardation, yet the FW-H mean is 0.35 times the estimate. On the axis $M_r = 0$ and $\dot{\mathbf{n}} \cdot \hat{\mathbf{x}} = 0$ for a rotation about z, so the FW-H integrand there is that of a fixed surface, and the surface motion cannot explain the axial gaps. The first suspect is the input to the acoustic step (pressure scaling, surface normals, or the library output convention); it has not been tested.

Not verified. No mesh-independence study (one mesh), no second time step, no resolved boundary layer on the blades, the phase-2 Courant number is not available (its log is on the offline cluster), and nothing is compared with a measurement.

Table 3: FW-H output against the compact-dipole estimate (3) from the blade pressure force ($r = 1\,\mathrm{m}$; script `paper/scripts/dipole_check.py`).

Component, microphone	force (mN)	compact dipole	FW-H	FW-H − dipole (dB)
Mean, downstream axis	44.4	3.53 mPa	1.24 mPa	-9.1
25 Hz, radial	0.77	7.5 dB	21.2 dB	+13.7
BPF (124.5 Hz), downstream axis	0.36	8.1 dB	-9.8 dB	-17.9

6 Results

6.1 Aerodynamics

Averaged over the last 50 ms of the force record, the rotor gives 43.3 mN of thrust and 1.023 mN m of torque, that is an aerodynamic power $C\Omega = 0.162\,\mathrm{W}$. The flow rate through a cut 60 mm downstream of the rotor is 30.1 L/s (108 m^3/h). The blades drive an annular jet along the duct wall, with a slow recirculating wake behind the hub (Figure 3).

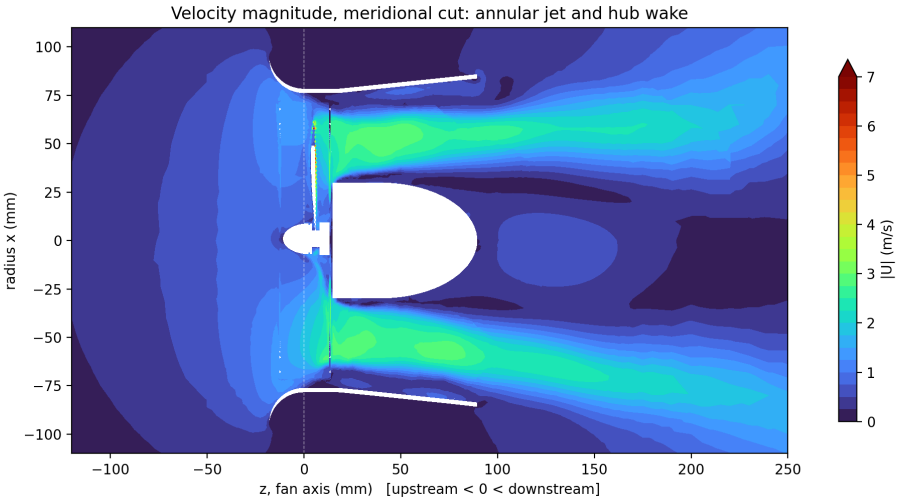


Figure 3: Velocity magnitude in the meridional plane at the last time step.

6.2 Spectra

Figure 4 shows the averaged spectra relative to each microphone's floor, and Table 4 the tone levels. The BPF stands 20 to 28 dB above the floor on all four microphones; its second harmonic appears on every microphone and its third on all but the radial one. The once-per-revolution tone dominates in the rotor plane and is absent on the axis. Neither tone should reach the axis from a perfectly symmetric rotor: a steady rotating loading of five identical blades radiates only modes $m = 5k$, and rotor–arm interaction with three arms gives $m = 5k - 3s$, so the axisymmetric mode $m = 0$ that radiates on the axis first appears at 3×BPF [2]. The on-axis BPF, like the 25 Hz tone, therefore implies a loss of five-fold symmetry in the computed loading.

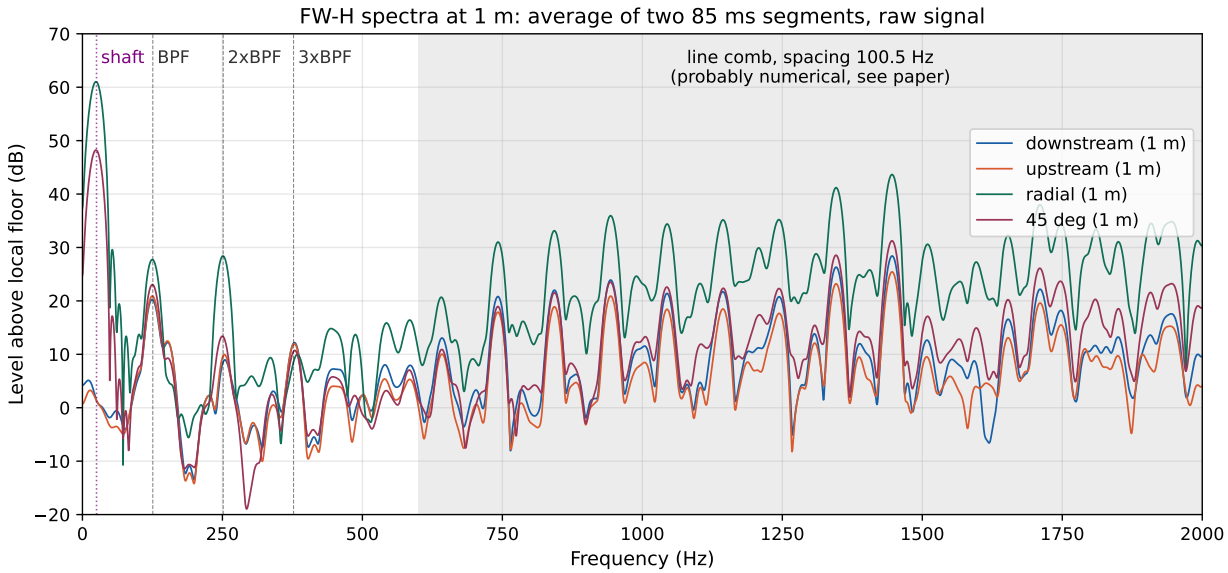


Figure 4: FW-H spectra at 1 m relative to each microphone's floor (140–240 Hz): average of the two 85 ms segments, Hann window, raw signal. Shaded: the line comb of Section 6.4.

Table 4: Tone levels at 1 m (dB re 20 µPa, raw signal). Absolute levels are not reliable (Section 5); only comparisons between microphones and tone-to-floor ratios are used.

Microphone	25 Hz	BPF	floor (140–240 Hz)	BPF − floor
Downstream (axis)	−26.7 (floor)	−9.8	−30.0	20.3
Upstream (axis)	−27.5 (floor)	−9.2	−30.1	20.9
Radial	21.2	−12.1	−39.8	27.7
45° downstream	18.1	−7.1	−30.1	23.0

6.3 Once-per-revolution tone

The net pressure force on the blades has an in-plane component of 0.77 mN that rotates with the rotor: F_x and F_y oscillate at 25.15 Hz with equal amplitudes. As shown in Section 5, the 25 Hz directivity is that of this rotating force. Whether the imbalance comes from the STL and mesh (the surfaces are faceted) or from the flow was not established.

6.4 High-frequency line comb

Above 600 Hz the spectra contain lines at 642, 742, 843, 943, 1044, 1144, 1244, 1346 and 1 446 Hz, spaced 100.5 Hz, up to 44 dB above the floor on the radial microphone; on the axial microphones they carry about 90 % of the signal variance. Folding the envelope of the high-passed signals on a 10 ms period shows bursts in the axial blade force about 1.4 ms after each multiple of 10 ms, and the same bursts in the FW-H signal 2.8 ms later, the propagation time over 1 m (Figure 5). The period matches the field write interval (10 ms) and also a quarter revolution (9.94 ms); the two folds are equally sharp, and 170 ms of data cannot separate them. No physical source at four events per revolution is expected for five blades and three arms, so the comb is most probably numerical. A rerun with a different write interval would decide. It lies well above the tones studied here and is excluded from the results.

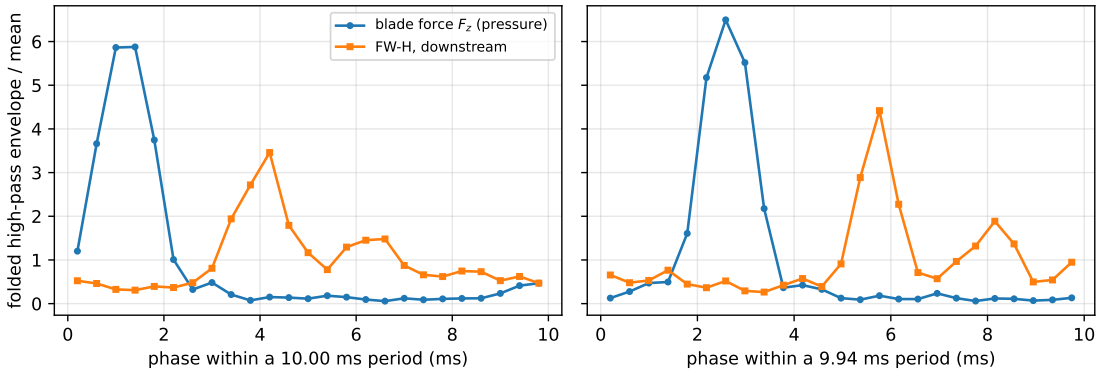


Figure 5: Envelope of the high-passed (> 600 Hz) axial blade force and downstream FW-H signal, folded on 10.00 ms (left) and 9.94 ms (right). Script `paper/scripts/comb_check.py`.

7 Discussion and limitations

The flow shows the expected annular jet. In the acoustic output, the frequencies and directivities are consistent with the blade forces, while the amplitudes are not; the on-axis BPF and the 25 Hz tone both imply a loss of five-fold symmetry, and the line comb above 600 Hz is probably numerical. The limits are:

- *Amplitudes.* The FW-H levels differ from the force-based estimates by −9 to +14 dB (Table 3).
- *Resolution.* Two 85 ms segments give 11.8 Hz. A contiguous 0.4 s record would take more than 100 h on the 1.95 M-cell mesh (the measured cost is 16.5 s per step on the 0.91 M-cell mesh), and the cluster holding the restart data was offline in October 2026.
- *Numerical accuracy.* One mesh, one time step, first-order time integration (227 steps per BPF period in phase 2), no prism layers, 49 827 concave cells, and solver runs with floating-point trapping off.
- *Domain.* The lateral boundaries are 0.2 m from the axis, 2.6 duct radii; their effect on the jet and on the rotor loading was not tested.
- *Physics.* URANS resolves tonal noise only. Serrations act mainly on broadband noise [3, 4], and no smooth-blade reference was run, so this study does not measure their effect. The FW-H surface covers the blades only.
- *No measurement.* Thrust, flow rate and spectra are not compared with tests.

8 Conclusion and future work

A sliding-mesh URANS and FW-H chain gives 43.3 mN of thrust and 108 m^3/h at 1 509 rpm for the serrated ducted fan, with a BPF tone and harmonics on all microphones and a once-per-revolution tone confined to the rotor plane. Checking the FW-H output against the integrated blade force validated its directivities and showed that its amplitudes are not reliable. In order of priority: run the acoustic step on a case with a known answer (a stationary or rotating compact dipole) to locate the amplitude error; rerun with a different field-write interval to test the line comb; record a contiguous signal of at least 0.4 s; add two meshes and a second time step; and run a smooth-blade reference rotor before drawing any conclusion on the serrations.

Reproducibility statement

Case files: `openfoam_case/` (phase 1 `system/controlDict`, phase 2 `system/controlDict.phase2`, mesh `system/snappyHexMeshDict`); the STL geometry is not published. The published records were computed on the cluster with OpenFOAM v2506 and libAcoustics; the earlier 0.91 M-cell run used OpenFOAM v2406 under WSL. From the repository root (Python 3, numpy, scipy, matplotlib; seconds per script): `postprocessing/acoustics/post_acoustique_combine.py` (Table 4, Figures 4 and 6), `postprocessing/aero/perf_aero.py` (thrust, torque, power, Figure 2), `paper/scripts/dipole_check.py` (Table 3), `paper/scripts/comb_check.py` (Figure 5). The flow rate and Figure 3 need ParaView slices that are not published (`postprocessing/rendering/render_flow_2d.py`). The paper is rebuilt with `paper/build_paper.sh`.

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A Audit of the first acoustic analysis

The first analysis of these records concatenated the two segments and applied a 5-point median filter against numerical spikes. The concatenation joins segments 15 ms apart, which puts a phase jump of $360° \times 15\,\text{ms} \times 25.15\,\text{Hz} \approx 136°$ into the once-per-revolution component and splits its peak. The segments contain few outlier samples (at most 4 beyond 5 robust standard deviations), and the filter raises the floor by 9 to 24 dB, hides the BPF harmonics and moves the radial BPF level by about 4 dB (Figure 6). Both steps were dropped.

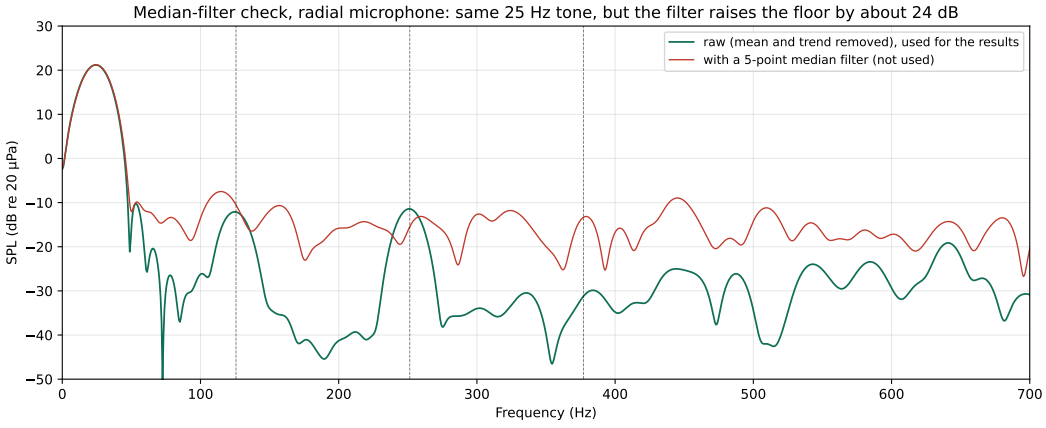


Figure 6: Radial microphone, average of the two segments: raw signal (used) and 5-point median filter (not used).